

Product Requirement Document

Product Name

Student Result Management System

Problem Statement

Current methods for managing student marks are often manual, error-prone, and time-consuming, leading to inefficiencies in result processing, report generation, and performance analysis. This makes it difficult for educational institutions to provide timely and accurate feedback to students and stakeholders, and to identify areas for academic improvement.

Goals

- To provide a centralized and secure system for storing and managing student academic results.
- To automate the process of generating various student performance reports (e.g., grade cards, subject-wise reports).
- To enable educators and administrators to analyze student performance trends effectively.
- To improve the accuracy and efficiency of result management operations.
- To provide an intuitive interface for users to interact with the system.

User Personas

- Teacher: Responsible for entering marks, viewing student profiles, and generating basic reports for their classes.
- Administrator/Registrar: Manages student data, configures grading systems, generates comprehensive reports, and oversees system operations.
- Student: Views their own marks, progress reports, and academic performance history.

Functional Requirements

- User Management: Ability to create, update, and delete user accounts (Teachers, Administrators, Students); Role-based access control.
- Student Management: Ability to add, edit, and delete student records (student ID, name, class, etc.); Ability to assign students to classes/courses.

- **Course/Subject Management:** Ability to add, edit, and delete courses/subjects; Ability to assign teachers to courses.
- **Marks Entry:** Ability for teachers to enter marks for students for specific subjects/assessments; Support for different grading schemes (e.g., numerical, letter grades).
- **Result Processing:** Automatic calculation of total marks, percentages, and grades; Ability to define and apply grading rules.
- **Report Generation:** Generate individual student grade cards/report cards; Generate class-wise result summaries; Generate subject-wise performance reports; Generate performance trend reports over time; Export reports in various formats (e.g., PDF, Excel).
- **Performance Analysis:** Dashboard for administrators to view overall academic performance; Tools for identifying top performers and students needing intervention.
- **Data Security:** Secure storage of student data; Audit trails for data modifications.

Non Functional Requirements

- **Performance:** The system should process and display results for 1000 students within 5 seconds; Report generation should be completed within 10 seconds for a class of 60 students.
- **Security:** All user data and marks must be encrypted at rest and in transit; The system must be protected against common web vulnerabilities (e.g., SQL injection, XSS); Strict access control based on user roles.
- **Usability:** The user interface should be intuitive and easy to navigate for all user types; Clear error messages and help documentation should be provided.
- **Reliability:** The system should have an uptime of 99.9%; Data backup and recovery mechanisms must be in place.
- **Scalability:** The system should be able to support up to 10,000 students and 500 teachers initially, with potential for future expansion.
- **Maintainability:** The codebase should be well-documented and follow best coding practices.

Constraints

- Must be developed using open-source technologies.
- Deployment target is a cloud-based infrastructure.
- Project completion deadline is 6 months from the start date.
- Budget limitation for third-party integrations.

Assumptions

- Accurate and timely input of marks by teachers will be ensured.
- Existing student data will be available in a structured format for initial migration.
- Users will have basic computer literacy to operate the system.
- Necessary infrastructure (servers, network) will be provided.

Out Of Scope

- Student attendance tracking.
- Timetable generation.
- Financial/fee management.
- Library management.
- Communication features (e.g., email/SMS notifications directly from the system, beyond basic system alerts).
- Admission process management.